Idea Proposal

Rhinoplasty

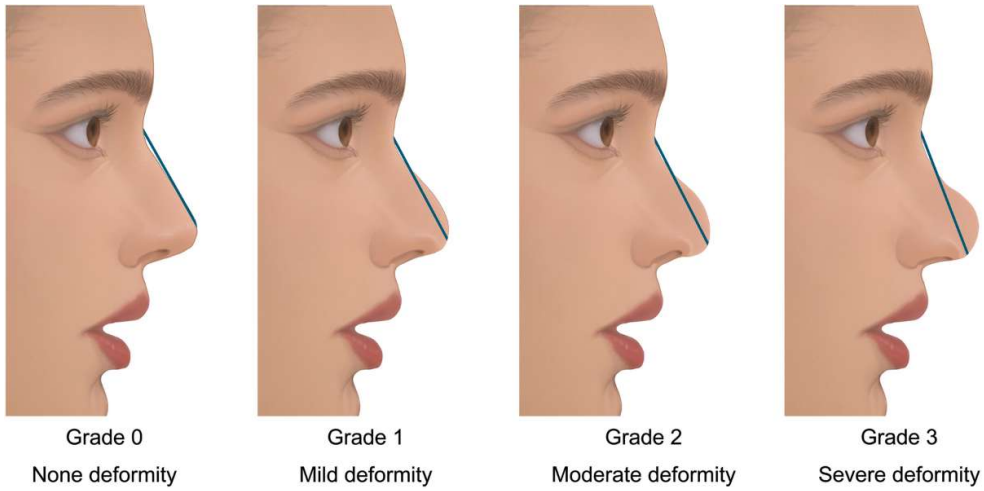
Subjects

1. Pollybeak Deformity

2. Diced Cartilage Fascia Grafting

3. Postoperative dynamic and traumatic nasal response

Pollybeak Deformity



Two types of deformity

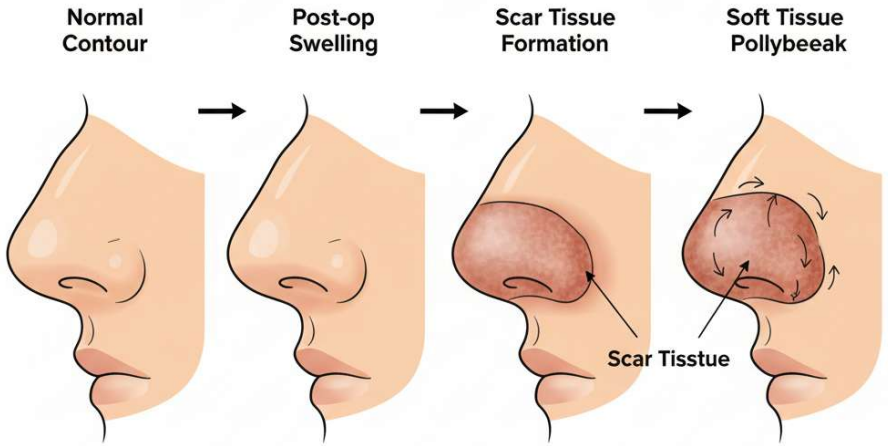
➔Cartilaginous Pollybeak

It results from insufficient cartilage removal during surgery.

➔Soft Tissue Pollybeak

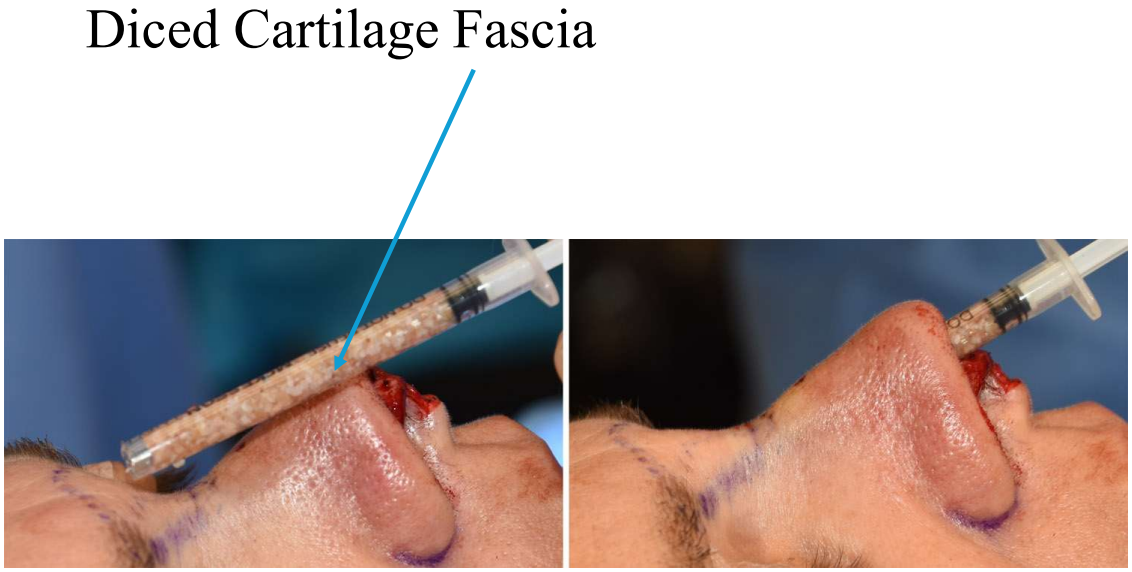
It is caused by excessive swelling or scar tissue formation.

How to analysis Soft Tissue Pollybeak



1. Different nonlinear material properties are assigned to normal skin and scar tissue.
2. Simulate the contraction forces of the scar tissue over time to predict how the soft tissue pulls on the underlying cartilaginous structures.
3. Be able to predict the final change in the nasal contour and potential for a pollybeak deformity to occur.

Diced Cartilage Fascia Grafting



A rhinoplasty technique where a surgeon places a patient's own finely diced cartilage, wrapped in a thin layer of fibrous tissue (fascia), into the nose to create a soft, natural-looking contour.

How to analysis

1. Build a 3D model of the nasal dorsum structure containing the DCF graft.
2. Analyze the stress and strain on the graft and surrounding tissues by simulating various *facial expressions* (smiling ,frowning, etc.) and *external loads* (wearing glasses)
3. Identify areas of stress concentration within the graft and at the graft-surrounding tissue interface to predict vulnerable points where long-term absorption or deformation might occur

-> It helps determine the optimal shape and volume of the graft before surgery

Postoperative dynamic and traumatic nasal response

After Surgery

The presence of scar tissue and weakened cartilage structures ⇨ Need a quantitative understanding of the nose's mechanical failure threshold ⇨ Under the damage from everyday (minor bumps, pressure from wearing glasses)

How to analysis



⇨ Analyze using FEA and CFD